Centre of gravity :- Module - 4

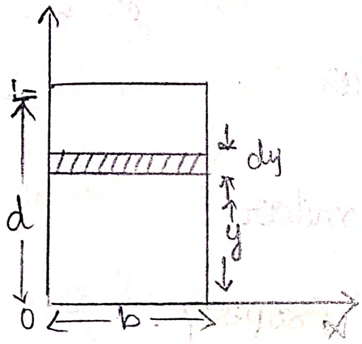
Centre of gravity of a body is a point through which the whole weight of the body acts. It is represented by CG or G.

There is only one CG for any position of the body.

Centroid :- Centroid of an area is a point at which the whole area of the plane fig. is assumed to be connectrated. it is represented by CG or G.

Location of centroid for regular figures from first principle.

(a) Rectangle



Let us consider a rectangle of weidth b and depth d as shown in fig. Consider a small strip of depth dy at a distance y from x-axis.

$A\bar{x} = \int ax = \Sigma ax$ — ①

$A\bar{Y} = \int ay = \Sigma ay$ — ②

Consider eq ②,

A = Area of whole figure = $b.d$

$\bar{Y}$ = Verticle centroid distance of rectangle from x-axis. = $\bar{y}$

a = Area of elemental strip = $b.dy$.

y = Centroid distance of elemental strip from x-axis = y.

$\therefore$ eq^n ② $\Rightarrow A\bar{y} = \int ay$.

$b.d\,\bar{y} = \int_0^d b\,dy\, y$.

$\cancel{b}d\bar{y} = \cancel{b}\int_0^d y.dy$

$\bar{y} = \frac{1}{d}\left[\frac{y^2}{2}\right]_0^d \Rightarrow \bar{y} = \frac{1}{\cancel{d}} \times \frac{1}{2} \times [d^{\cancel{2}} - 0^2] \Rightarrow \bar{y} = \frac{d}{2}$

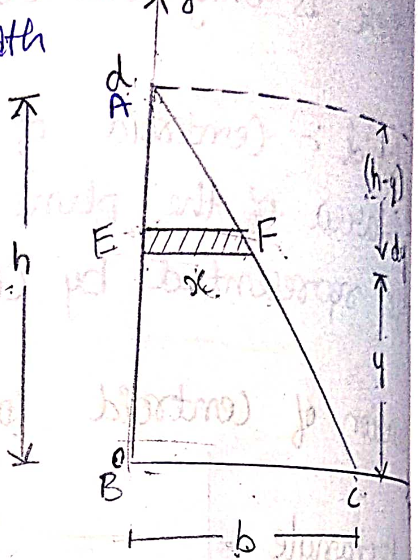
$\bar{x} = \frac{b}{2}$

$$\frac{1}{d} \times \frac{1}{2} \times [d^2 - 0^2]$$

$$\boxed{\bar{y} = \frac{d}{2}}$$

$$\boxed{\bar{x} = \frac{b}{2}}$$

* **Triangle** :- Let us consider a Δ^{le} of width b and a height h. as shown in fig.



Consider a small strip EF having thickness dy at a distance y from bottom x-axis.

Using the property of similar triangle we have,

Δ^{le} ABC and Δ^{le} AEF are similar

$$\frac{x}{b} = \frac{h-y}{h} \Rightarrow \text{Similar } \Delta^{le} \text{ Property.}$$

$$x = b\left(1 - \frac{y}{h}\right)$$

W.R.T.

$$A\bar{X} = \int ax = \Sigma ax \quad -(1)$$

$$A\bar{Y} = \int ay = \Sigma ay \quad -(2)$$

Consider eq^n (2) $\Rightarrow A = \frac{1}{2} b.h$

- $A =$ Area of triangle $= \frac{1}{2} \times b \times h$.
- $\bar{Y}$ = verticle centroid distance from x-axis. $= \bar{Y}$
- $a =$ Area of elemental strip. $x.dy$.

~~x.dy~~ = • $a = b\left(1 - \frac{y}{h}\right) dy$. , $y = y$

- $y = y$ = vertical centroid distance of the strip from bottom x-axis. $= y$.

Eq② $\Rightarrow A\bar{y} = \int ay$

$\frac{1}{2} bh \cdot \bar{y} = \int_0^h b(1 - y/h) \cdot dy \cdot y$

$\frac{1}{2} \cancel{b} h \bar{y} = \cancel{b} \int_0^h (y - y^2/h)\, dy$

$\bar{y} = \frac{2}{h}\left[\frac{y^2}{2} - \frac{1}{h} \cdot \frac{y^3}{3}\right]_0^h$

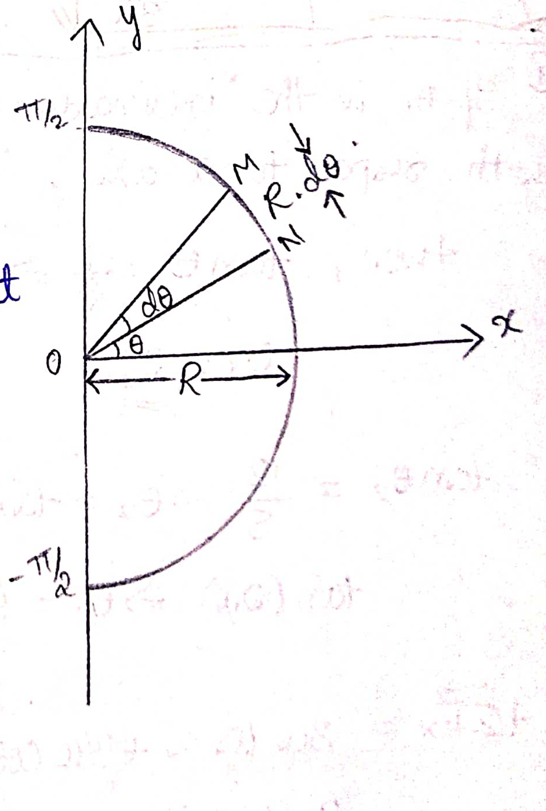
$= \left[\frac{2}{h} \frac{h^2}{2} - \frac{1}{\cancel{h}} \frac{\cancel{h^3}}{3}\right]$

$= \frac{2}{h}\left[\frac{3h^2 - 2h^2}{6}\right] \Rightarrow \frac{\cancel{2}}{\cancel{h}}\left[\frac{\cancel{h^2}}{\cancel{6}_3}\right]$

$\boxed{\bar{y} = \frac{h}{3}}$

Similarly, $\boxed{\bar{x} = \frac{b}{3}}$

(C) <u>Semi circle</u> :-



Consider a semicircle of radius R as shown in fig.

Consider an elemental area OMN at an angle "θ" w.r.to x-axis making an angle "$d\theta$".

W.K.T.

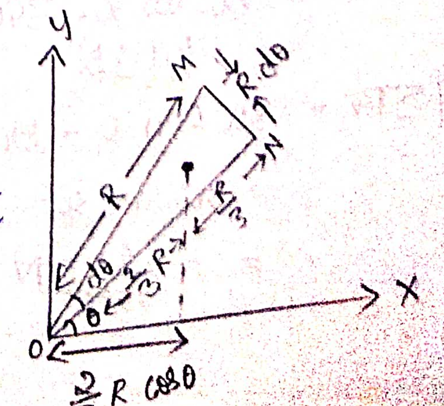
$A.\bar{x} = \int ax = \Sigma ax$ —①

$A = \frac{\pi R^2}{2}$

$\bar{x} = \bar{x}$

$a = \frac{1}{2} R \cdot R \cdot d\theta = \frac{R^2}{2} d\theta$

x = Horizontal centroid distance of element from y-axis $= \frac{2}{3} R \cdot \cos\theta$.



Eq^n ① $\Rightarrow$

$$A\bar{X} = \int a x$$

$$\frac{\pi R^2}{2}\bar{x} = \int_{-\pi/2}^{\pi/2} \frac{R^2}{2} d\theta \cdot \frac{2}{3} R\cos\theta$$

$$\frac{\pi R^2}{2}\bar{x} = \frac{R^3}{3}\int_{-\pi/2}^{\pi/2} \cos\theta \cdot d\theta$$

$$\bar{X} = \frac{2R}{3\pi}\left[\sin\theta\right]_{-\pi/2}^{\pi/2}$$

$$\bar{X} = \frac{2R}{3\pi}\left[\sin\pi/2 - \sin(-\pi/2)\right]$$

$$\bar{x} = \frac{2R}{3\pi}[1+1] \Rightarrow \frac{4R}{3\pi}$$

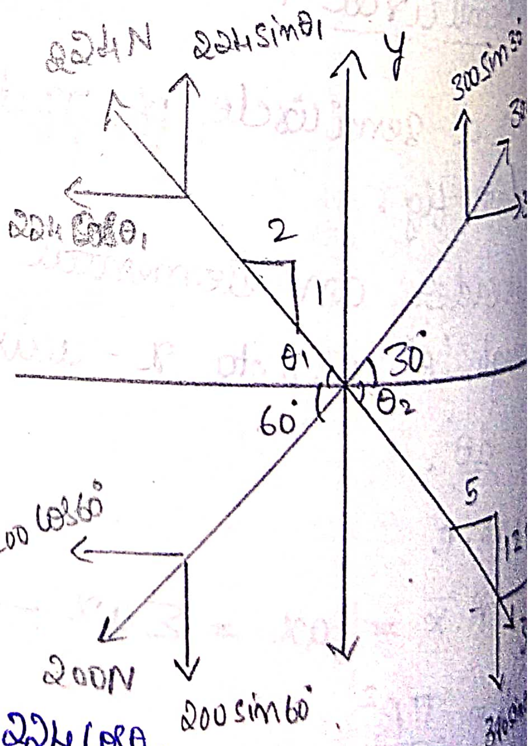
where $\boxed{\bar{x} = \frac{4R}{3\pi}}$

---

① If θ_1 is the inclination of 224N force with respect to x-axis,

then, $\tan\theta_1 = \frac{1}{2} \Rightarrow \theta_1 = \tan^{-1}\left(\frac{1}{2}\right)$

$\theta_1 = 26.5^\circ$



$\tan\theta_2 = \frac{12}{5} \Rightarrow \theta_2 = \tan^{-1}\left(\frac{12}{5}\right)$

$\tan^{-1}(2.4) \Rightarrow \theta_2 = 67.3^\circ$

$\overset{+}{\rightarrow}\Sigma F_x = 300\cos 30^\circ + 390\cos\theta_2 + 200\cos 60^\circ - 224\cos\theta_1$

$= 300\cos 30^\circ + 390\cos 67.3 - 200\cos 60^\circ - 224\cos 26.5$

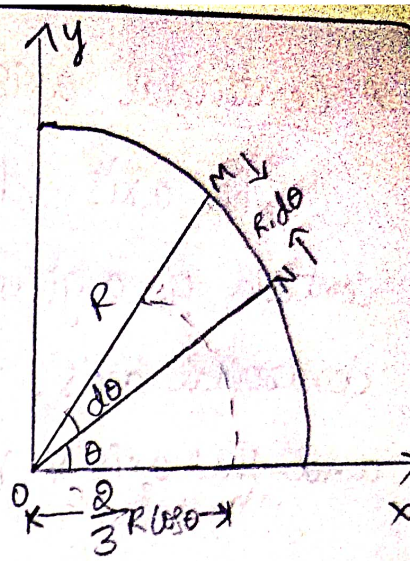
$= 109.8N$

$+\uparrow\Sigma F_y = 300\sin 30^\circ - 390\sin\theta_2 - 200\sin 60^\circ + 224\sin\theta_1$

$= 300\sin 30^\circ - 390\sin 67.3 - 200\sin 60^\circ + 224\sin 26.5$

$= -283.0\,N$

* Quandrant of a circle :-



W.K.T. $A\bar{X} = \int ax = \sum ax$ — ①

$A\bar{Y} = \int ay = \sum ay$ — ②

Consider a quandrant of a circle of radius R as shown in fig. consider an elemental area OMN having an angle dθ making an angle θ with respect to x-axis.

~~W.K.T.~~

Consider Eqn ①,

$$A = \frac{\pi R^2}{4}$$

$$\bar{X} = \bar{X}$$

$$a = \frac{1}{2} R . R d\theta = \frac{R^2}{2} d\theta$$

$$x = \frac{2}{3} R \cos\theta$$

Eq ① $\Rightarrow$ $A\bar{X} = \int ax$

$$\frac{\pi R^2}{4} \bar{X} = \int_0^{\pi/2} \frac{R^2}{2} d\theta . \frac{2}{3} R \cos\theta$$

$$\frac{\pi \cancel{R^2}}{4} \bar{X} = \frac{R^{\cancel{3}}}{3} \int_0^{\pi/2} \cos\theta . d\theta$$

$$\bar{X} = \frac{4R}{3\pi} [\sin\theta]_0^{\pi/2}$$

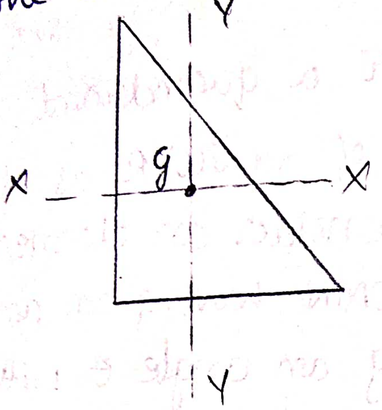
$$\bar{X} = \frac{4R}{3\pi} [\sin \pi/2 - \sin 0]$$

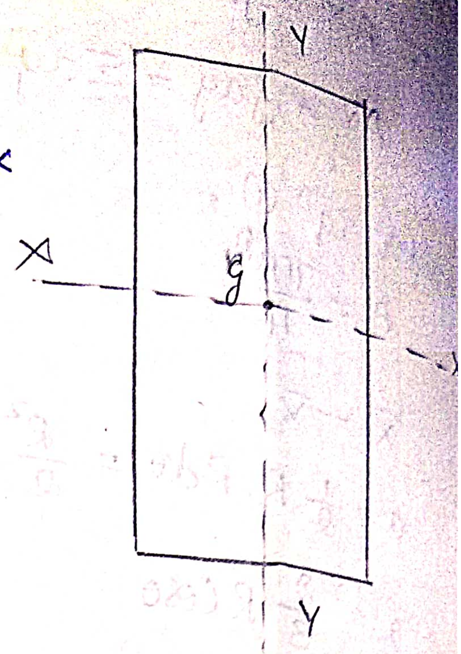
$$\boxed{\bar{X} = \frac{4R}{3\pi}}$$

similarly $\boxed{\bar{Y} = \frac{4R}{3\pi}}$

* Centroidal axis :-

The axis which passes through the centroid of a given figure is known as centroidal axis, such as the axis X–X and the axis Y–Y as shown in fig.

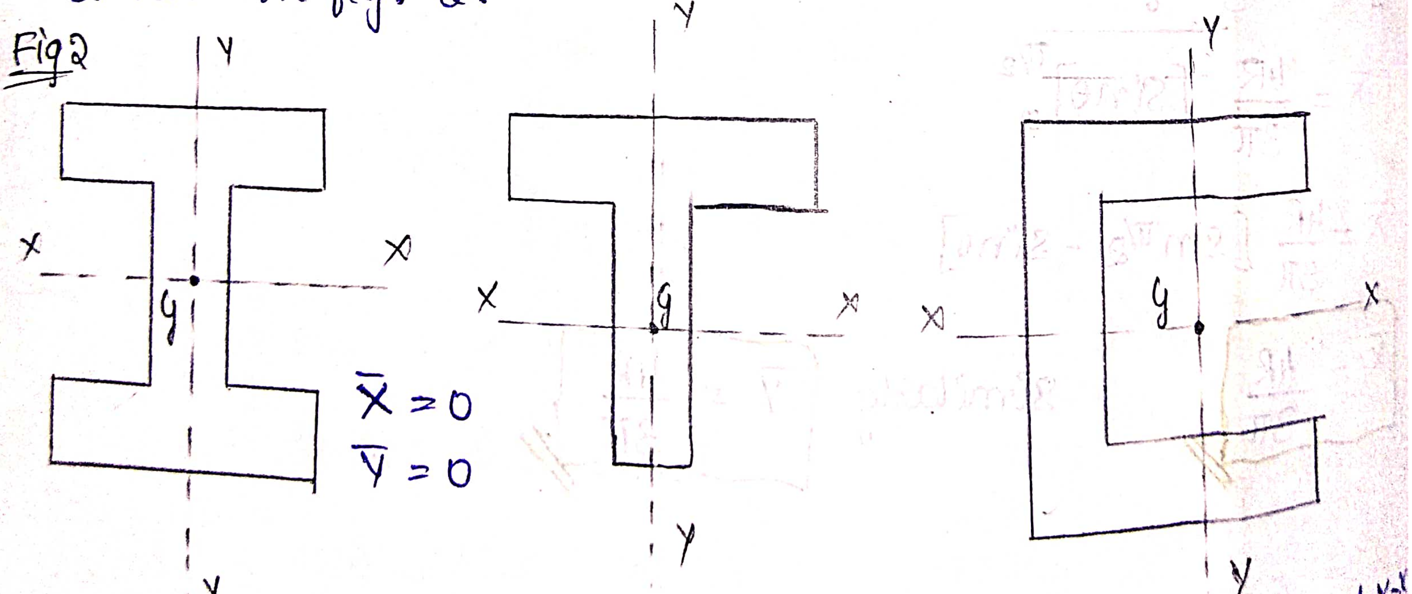




Symmetrical axis

It is the axis which divides the whole figure into equal parts such as the axis X–X and the axis Y–Y shown in fig. 2.

Fig 2

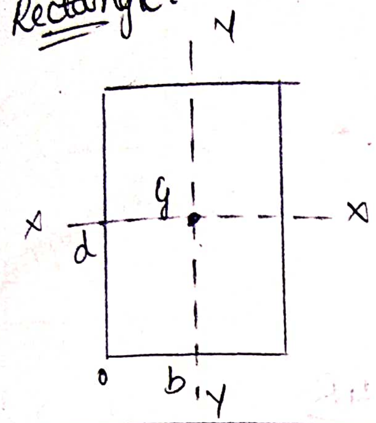
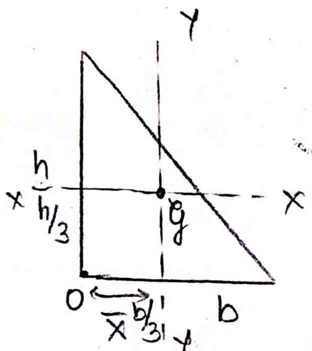
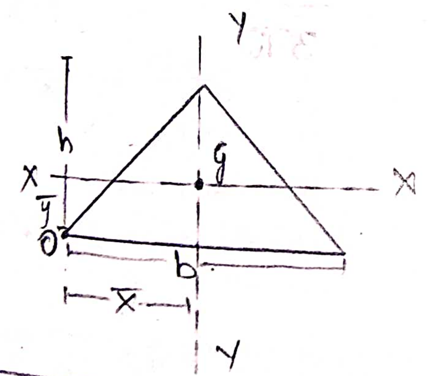
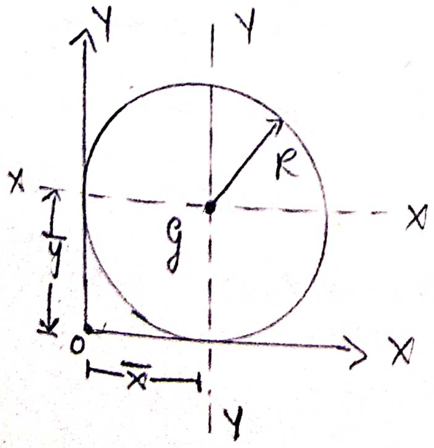


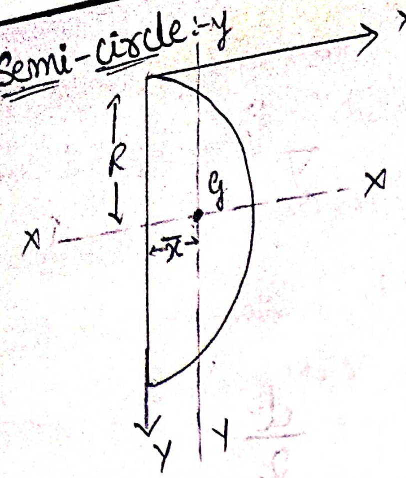
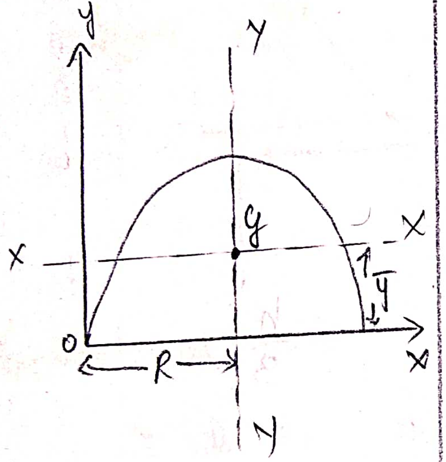
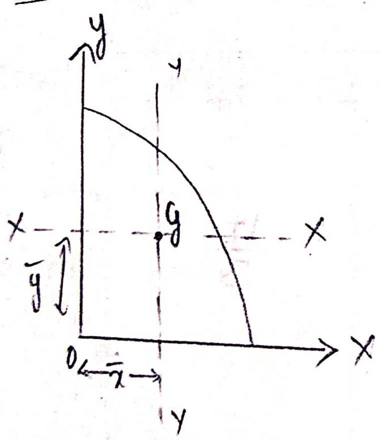
Symmetrical axis (Both)

Symmetrical about Y–Y axis ($\bar{x} = 0$)

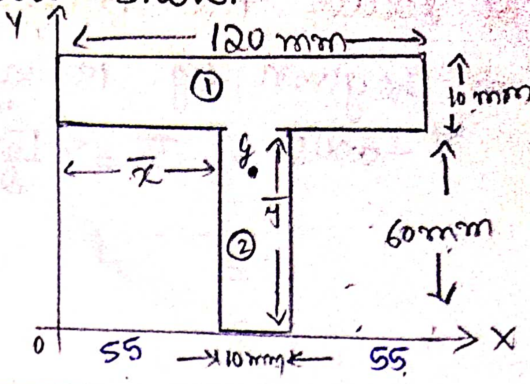
Symmetrical about X–X axis ($\bar{Y} = 0$).

Abstract / Summary :-

	Area	$\bar{X}$	$\bar{Y}$
Rectangle :- 	bd	$\frac{b}{2}$	$\frac{d}{2}$
Right angled Triangle :- 	$\frac{1}{2}bh$	$\frac{b}{3}$	$\frac{h}{3}$
Isoceles Triangle :- 	$\frac{1}{2}bh$	$\frac{b}{2}$	$\frac{h}{3}$
Circle :- 	πR^2	R	R

Semi-circle :- 	$\frac{\pi R^2}{2}$	$\frac{4R}{3\pi}$	$-R$
	$\frac{\pi R^2}{2}$	R	$\frac{4R}{3\pi}$
① * Quadrant of a circle :- 	$\frac{\pi R^2}{4}$	$\frac{4R}{3\pi}$	$\frac{4R}{3\pi}$

1. Determine the centroid of the figure shown



Component	Area a mm^2	Centroidal distance from y-axis x in mm	Centroidal distance from x-axis y in mm	ax in mm^3	ay in mm^3
Rectangle ①	$120 \times 10 =$ 1200	$\frac{120}{2} = 60$	$60 + \frac{10}{2} = 65$	72,000	78,000
Rectangle ②	$10 \times 60 =$ 600	$55 + \frac{10}{2} =$ 60	$\frac{60}{2} = 30$	36,000	18,000
	$\Sigma a = 1800$			$\Sigma ax =$ 1,08,000	$\Sigma ay =$ 96,000

W.K.T.

$$\bar{x} = \frac{\Sigma ax}{\Sigma a} \Rightarrow \frac{1,08,000}{1800} = 60 \text{ mm}$$

$\therefore \boxed{\bar{x} = 60 \text{ mm}}$

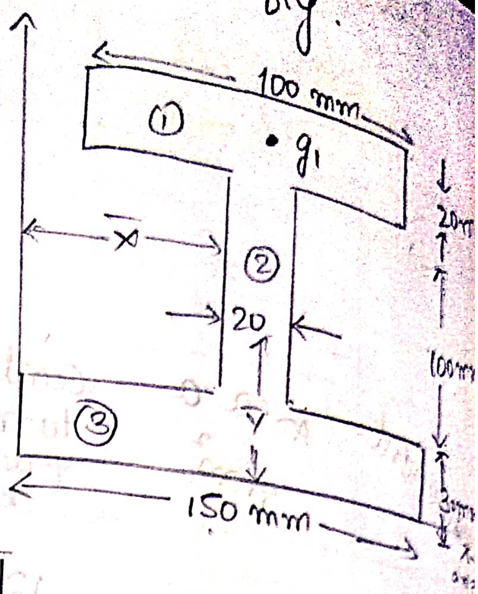
$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{96000}{1800} = 53.33 \text{mm}$$

$\therefore \boxed{\bar{Y} = 53.33 \text{ mm}}$

The given figure is symmetrical about the Y-axis, so $\bar{x}$ can be directly written as $\frac{120}{2} = 60$ mm.

2. Locate the centroid of i- section shown in fig.

⇒ The given fig. is symmetrical about y-axis, $\bar{x} = \frac{150}{2} = 75$ mm.



Component	a mm^2	y mm	ay mm^3
Rectangle ①	100×20 $= 2000$	$30 + 100 + \frac{20}{2}$ $= 140$	2,80,000.
Rectangle ②	20×100 $= 2000$	$30 + \frac{100}{2}$ $= 80$	1,60,000
Rectangle ③	150×30 $= 4500$	$\frac{30}{2} = 15$	67500
	$\Sigma a = 8500$		$\Sigma ay = 5,07,500$

W.K.T.

$$\bar{y} = \frac{\Sigma ay}{\Sigma a}$$

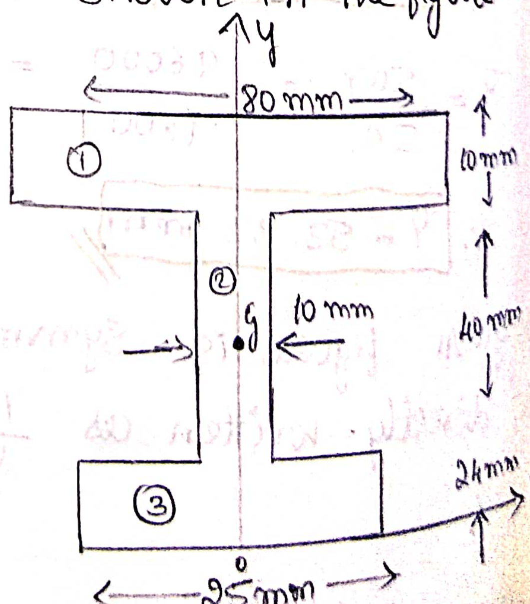
$$= \frac{5,07,500}{8500}$$

$$\boxed{\bar{y} = 59.70 \text{ mm}}$$

3. Locate the centroid of I- section shown in the figure with respect to point - o.

⇒ Due to symmetry centroid must lie on y-axis.

$\therefore \bar{x} = 0$.



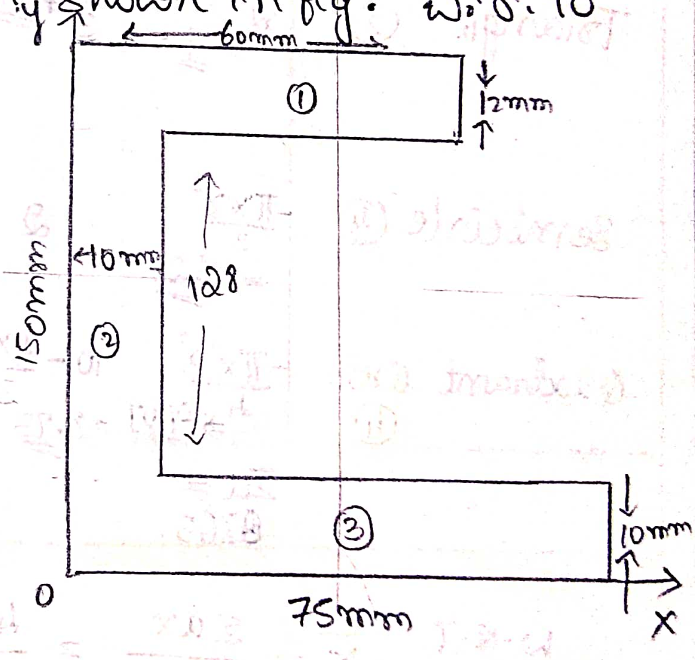
Component	a mm^2	y mm	ay mm^3
Rectangle ①	$80 \times 10 = 800$	$24+40+\frac{10}{2} = 69$	55200
Rectangle ②	$10 \times 40 = 400$	$24+\frac{40}{2} = 44$	17600
Rectangle ③	25×24 600	$\frac{24}{2} = 12$	7200
	$\Sigma a = 1800$		$\Sigma ay = 80{,}000$

W.K.T.

$$\bar{y} = \frac{\Sigma ay}{\Sigma a} = \frac{80{,}000}{1800}$$

$$\boxed{\bar{y} = 44.4\ mm}$$

4. Locate the centroid of C-section shown in fig. w.r.to point "o".



Component	a mm^2	x mm	y mm	ax mm^3	ay mm^2
Rectangle ①	$60 \times 12 = 720$	$\frac{60}{2} = 30$	$10+128+\frac{12}{2} \Rightarrow 144$	21600	103680
Rectangle ②	$10 \times 128 = 1280$	$\frac{10}{2} = 5$	$10+\frac{128}{2} = 74$	6400	94720
Rectangle ③	$75 \times 10 = 750$	$\frac{75}{2} = 37.5$	$\frac{10}{2} = 5$	28,125	3750
	$\Sigma a = 2750$			$\Sigma ax = 56{,}125$	$\Sigma ay = 202{,}150$

W.K.T.

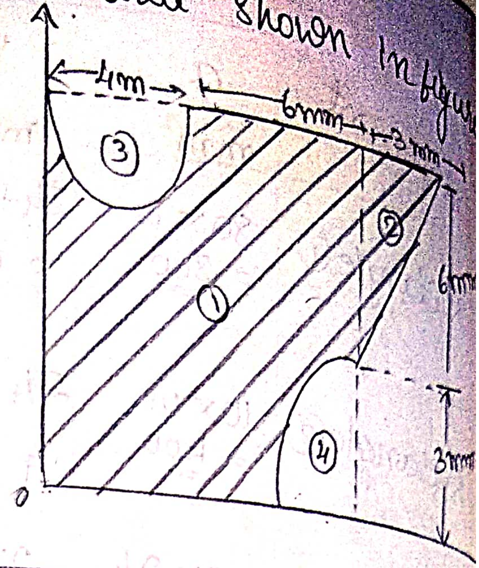
$$\bar{X} = \frac{\Sigma ax}{\Sigma a} = \frac{56{,}125}{2750}$$

$$= 20.40\ mm$$

$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{202{,}150}{2750}$$

$$= 73.50\ mm.$$

5. Determine the centroid of the shaded area shown in figure

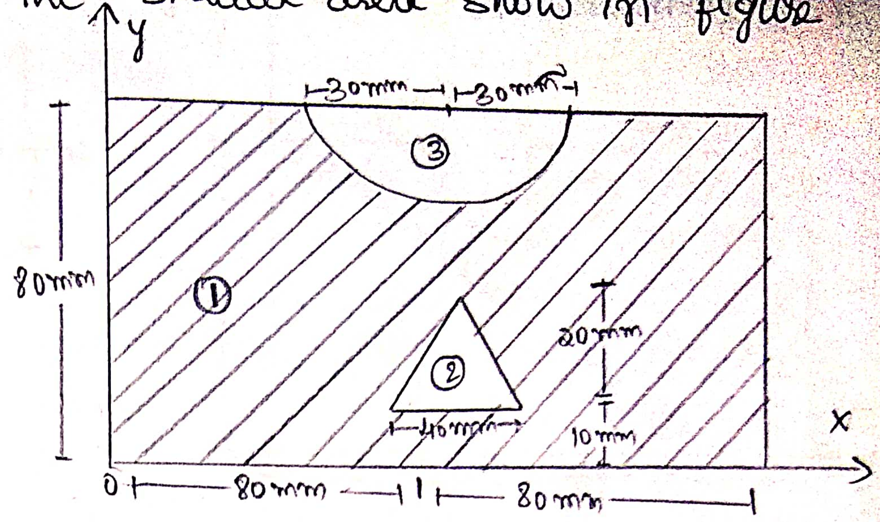


Component	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Rectangle ①	$10\times9 = 90$	$\frac{10}{2} = 5$	$\frac{9}{2} = 4.5$	450	405
Triangle ②	$\frac{1}{2}\times3\times6 = 9$	$\frac{3}{3}+10 = 11$	$9 - \frac{6}{3} = 7$	99	63
Semicircle ③	$-\frac{\pi\times2^2}{2} = -6.28$	2	$9 - \frac{4\times2}{3\pi} = 8.15$	-12.56	-51.182
Quadrant circle ④	$-\frac{\pi\times3^2}{4} = -7.07$	$10 - \frac{4\times3}{3\pi} = 8.73$	$\frac{4\times3}{3\pi} = 1.27$	-61.72	-8.978
	$\Sigma a = 85.65$.			$\Sigma ax = 474.72$	$\Sigma ay = 407.84$

W.K.T. $\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{474.72}{85.65} = 102.09$

$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{407.84}{85.65} = 87.70$

6. Locate the centroid of the shaded area show in figure with respect to "O".



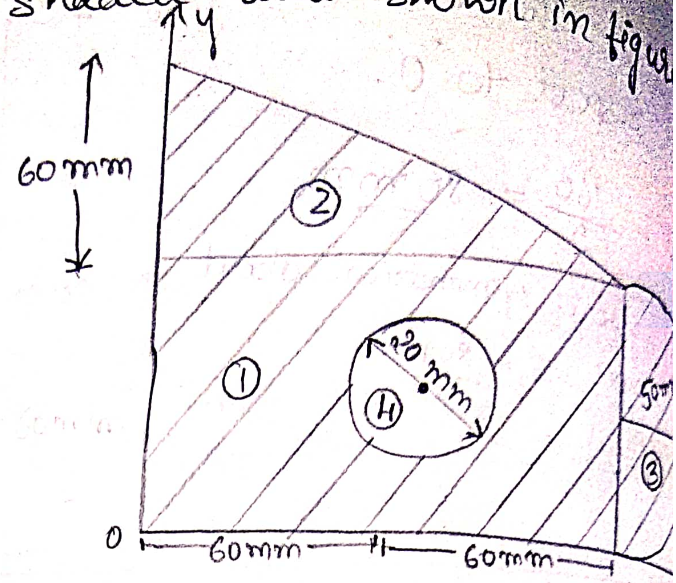
$\bar{x} = \frac{160}{2} = 80\,mm$

The given figure is symmetrical about y-axis hence . $\bar{x} = 80\,mm$

component	a mm^2	y mm	ay mm^3
Rectangle ①	$160 \times 80 = 12,800$	$\frac{80}{2} = 40$	512,000
Triangle ②	$-\frac{1}{2} \times 40 \times 20 = -400$	$\frac{20}{3} + 10 = 16.67$	6,668
semi-circle ③	$-\frac{\pi R^2}{2} = -\frac{3.14 \times 30^2}{2} = -1,413$	$80 - \frac{4R}{3\pi}$ $80 - \frac{4 \times 30}{3 \times 3.14}$ 67.7	-95,660
	$\Sigma a = 10,987$		$\Sigma ay = 409.672$

$\bar{y} = \frac{\Sigma ay}{\Sigma a} = \frac{409.672}{10,987} = 37.4$

5. 7. Determine the centroid of shaded area shown in figure.



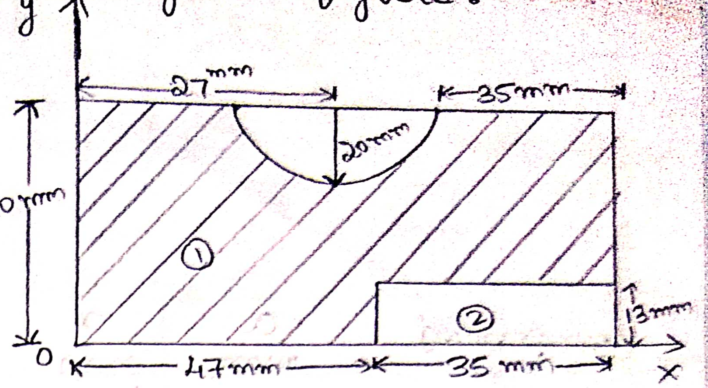
Component	a in mm^2	x in mm	y in mm	ax in mm^3	ay in mm^3
Rectangle ①	$120 \times 100 = 12000$	$\frac{120}{2} = 60$	$\frac{100}{2} = 50$	720,000	600,000
Triangle ②	$\frac{1}{2} \times 120 \times 60 = 3600$	$\frac{120}{3} = 40$	$100 + \frac{60}{3} = 120$	144,000	432,000
Semi-cirle ③	$\frac{\pi \times 50^2}{2} = 3925$	$120 + \frac{4 \times 50}{3\pi} = 141.2$	50	554,210	196,250
Circle ④	$-\pi \times 10^2 = -314$	60	50	-18,840	-15,700
	$\Sigma a = 19211$			$\Sigma ax = 1399,370$	$\Sigma ay = 1,212,550$

W.K.T.

$$\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{1399,370}{192.11} = 72.84 \text{ mm}$$

$$\bar{Y} = \frac{\Sigma aY}{\Sigma a} = \frac{1212550}{19211} = 63.11 \text{ mm}$$

8. Locate CG for shaded area for the given figure.
[Centroid gravity]



Component	a in mm^2	x in mm	y in mm	ax in mm^3	ay in mm^3
Rectangle ①	$82 \times 40 = 3280$	$\frac{82}{2} = 41$	$\frac{40}{2} = 20$	134,480	65,600
Rectangle ②	$-13 \times 35 = -455$	$47 + \frac{35}{2} = 64.5$	$\frac{13}{2} = 6.5$	−29,347.5	−2,957.5
Semi-circle ③	$-\frac{\pi \times 20^2}{2} = 628$	27	$40 - \frac{4 \times 20}{3\pi} = 31.51$	−16,956	−19,788.2
	$\Sigma a = 2197$			$\Sigma ax = 88176.5$	$\Sigma ay = 42860.5$

W.K.T.

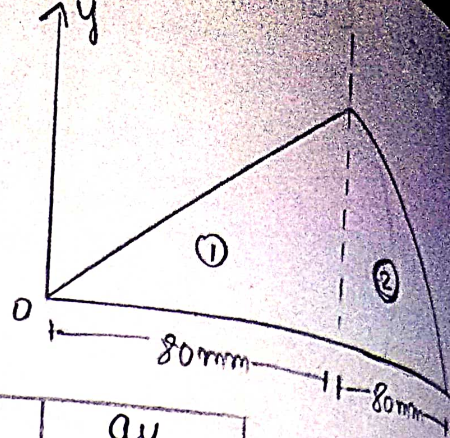
$$\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{88176.5}{2197} = 40.13 \text{ mm.}$$

$$\bar{y} = \frac{\Sigma ay}{\Sigma a} = \frac{42860.5}{2197} = 19.6 \text{ mm.}$$

$\bar{x} = 40.1$

$\bar{y} = 19.6$.

9. Determine the centroid of figure showed



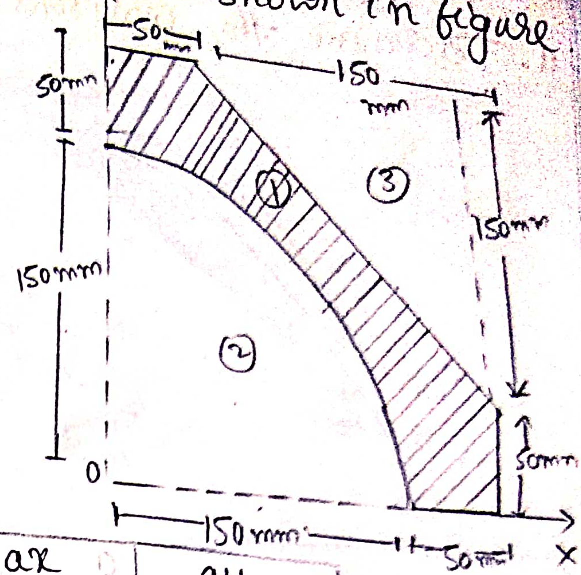
Component	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Triangle ①	$\frac{1}{2} \times 80 \times 80$ = 3,200	$\cancel{80}\,\frac{80}{3}$ = 53.3	$\frac{80}{3}$ = 26.6	170560	85,120
Quadrant of ~~(semi)~~ circle ②	$\frac{\cancel{3.14} \times 80^2}{\cancel{4}}$ 5024	$\frac{4R}{3\pi}+80$ $\frac{4 \times 80}{3 \times 3.14}$ = 113.97	$\frac{4R}{3\pi}$ = 33.97	572585.28	170,665.28
	Σa = 8224			Σax = 743241.28	Σay = 255,785.28

W.K.T.

$$\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{743241.28}{8224} = 90.37\ mm$$

$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{255,785.28}{8224} = 31.10\ mm.$$

10. Determine the centroid of the shaded area shown in figure



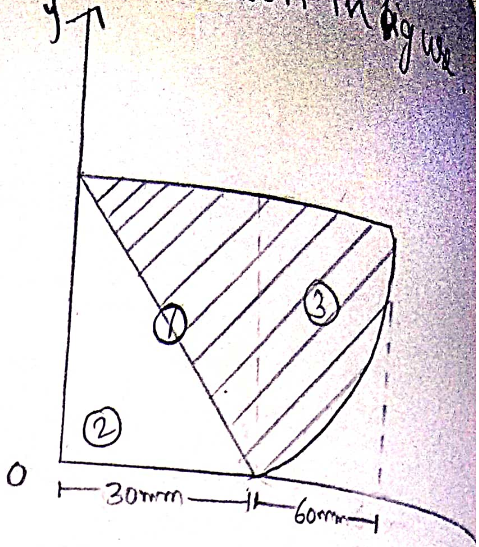
Component	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Rectangle ①	$200 \times 200 = 40{,}000$	$\frac{200}{2} = 100$	$\frac{200}{2} = 100$	4,000,000	4,000,000
Quadrant of a circle ②	$\frac{\pi \times 150^2}{4} = -17{,}662.5$	$\frac{4 \times 150}{3\pi} = 63.69$	$\frac{4 \times 150}{3\pi} = 63.69$	-1,124,924.6	-1,124,924.6
Triangle ③	$\frac{1}{2} \times 150 \times 150 = -11250$	$200 - \frac{150}{3} = 150$	$200 - \frac{150}{3} = 150$	-1,687,500	-1,687,500
	$\Sigma a = 11087.5$			$\Sigma ax = 1187575.4$	$\Sigma ay = 1187575.4$

W.K.T.

$$\bar{X} = \frac{\Sigma ax}{\Sigma a} = \frac{1187575.4}{11087.5} = 107.10$$

$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{1187575.4}{11087.5} = 107.10$$

9.11. Determine the centroid of shaded area shown in figure.



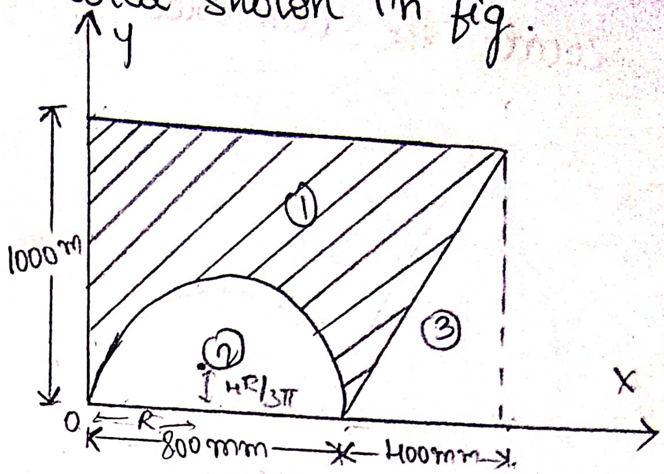
Components	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Rectangle 1	30×60 $= 1,800$	$\frac{30}{2} = 15$	$\frac{60}{2} = 30$	27,000	54,000
Triangle 2	$\frac{1}{2} \times 30 \times 60$ $= -900$	$\frac{30}{3} = +10$	$\frac{60}{3} = +20$	-9,000	-18,000
Quadrant of circle 3	$\frac{\pi \times 60^2}{4}$ $= 2826$	$30 + \frac{4(60)}{3 \times 3.14}$ $= 55.47$	$60 - \frac{4(60)}{3 \times 3.14}$ $= 34.53$	156,758.22	97581.78
	$\Sigma a = 3726$			$\Sigma ax =$ 174,758.22	$\Sigma ay =$ 133581.78

W.K.T.

$$\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{174,758.22}{3726} = 46.90$$

$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{133581.78}{3726} = 35.85$$

Locate the centroid of the shaded area shown in fig.



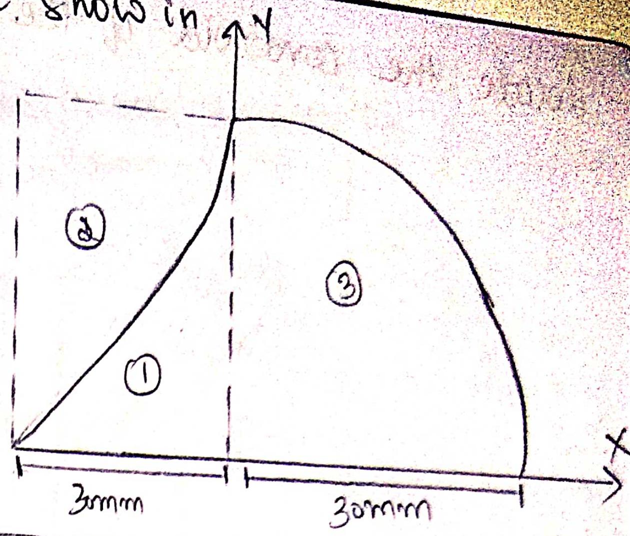
Component	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Rectangle ①	$1200 \times 1000 = 1200000$	$\frac{1200}{2} = 600$	$\frac{1000}{2} = 500$	720000000	600000000
Semi-circle ②	$-\frac{\pi \times 400^2}{2} = -251200$	$R = 400$	$\frac{4 \times 400}{3\pi} = 169.85$	-100,480,000	-42,666,320
Triangle ③	$\frac{1}{2} \times 400 \times 1000 = 200000$	$1200 - \frac{400}{3} = 1066.7$	$\frac{1000}{3}$ 333.3	-213,340,000	-66,660,000
	$\Sigma a =$ 748,800			$\Sigma ax =$ 406,180,000	$\Sigma ay =$ 490673,680

W.K.T.

$$\bar{x} = \frac{\Sigma ax}{\Sigma a} = \frac{406,180,000}{748,800} = 542.451$$

$$\bar{Y} = \frac{\Sigma ay}{\Sigma a} = \frac{490673,680}{748,800} = 655.28$$

Locate the centroid of the shaded area show in fig.



Component	a mm^2	x mm	y mm	ax mm^3	ay mm^3
Rectangle 1	30×30 $= 900$	-15	15	-13500	13500
Quadrant of circle 2	$\frac{\pi R^2}{4}$ $\Rightarrow \frac{3.14 \times 900}{4}$ 706.5	$\frac{4R}{3\pi}$ $\frac{4 \times 30}{3 \times 3.14}$ $= 12.7$	$\frac{4R}{3\pi}$ $\frac{4 \times 30}{3 \times 3.14}$ $= 12.7$	8972.55	8972.55
Quadrant of circle 3	$\frac{\pi R^2}{4}$ $= -706.5$	$30 - \frac{4R}{3\pi}$ $= 17.27$	$30 - \frac{4R}{3\pi}$ $= 17.27$	-12201.255	-12201.255
	$\Sigma a = 900$			$\Sigma ax = 10271.295$	$\Sigma ay = 10271.295$

$\Sigma \bar{x} = 8.4$

$\Sigma \bar{Y} = 11.4$